ECON 423/523-Lab-6

Sonan Memon

University of Oregon



Bad Controls

- Adding controls allows us to deal with omitted variable bias but adding bad controls can create its own problems.
- Bad controls are either highly correlated with independent variable of interest or the outcome variable.
- Bad controls can kill the variation of interest in the outcome variable or if correlated with indepedent variables of interest, create multi-collinearity.

Hypothesis Testing:

- **Confidence Interval:** A confidence interval is an interval estimator, constructed from sample data such that, under repeated random sampling, it contains the true population parameter with a probability $1 - \alpha$.
- **Rejection Region:** The rejection region is the set of values of a test statistic for which the null hypothesis H_0 is rejected. It is determined by the significance level α.
- **Type I Error (α):** A Type I error occurs when the null hypothesis H_0 is rejected even though it is true.

Hypothesis Testing:

- **Type II Error (β):** A Type II error occurs when the null hypothesis H_0 is not rejected even though it is false.
- **Statistical Power ($1 - \beta$):** Statistical power is the probability of correctly rejecting the null hypothesis when it is false.
- **p-value:** The p-value is the probability, assuming the null hypothesis H_0 is true, of observing a test statistic at least as extreme as the one actually observed.
- There is a trade-off between Type I and Type II errors: holding sample size, effect size, and the test statistic's distribution fixed, reducing α generally increases β.

Hypothesis Testing: Probabilities

- **Type I Error (α):**

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

- **Correct Non-Rejection ($1 - \alpha$):**

$$1 - \alpha = P(\text{Do not reject } H_0 \mid H_0 \text{ is true})$$

- **Type II Error (β):**

$$\beta = P(\text{Do not reject } H_0 \mid H_1 \text{ is true})$$

- **Statistical Power ($1 - \beta$):**

$$1 - \beta = P(\text{Reject } H_0 \mid H_1 \text{ is true})$$

P-Value:

- ϕ is CDF for standard, normal distribution.
- For instance, if $p = 0.045$, we can reject the null for all $\alpha \geq 0.045$ but not for $\alpha \in (0, 0.045)$.

$$z_{\text{obs}} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

$$p = 2P(Z \geq |z_{\text{obs}}| \mid H_0) = 2\left[1 - \Phi\left(|z_{\text{obs}}|\right)\right]$$

Hypothesis Testing: Decision Outcomes

Decision	H_0 **is True**	H_0 **is False**
Reject H_0	Type I Error (α)	Power ($1 - \beta$)
Fail to Reject H_0	Confidence Level ($1 - \alpha$)	Type II Error (β)

Instrumental Variables

- Instrumental variables have allowed many people to get tenure in the economics profession; they deal with problem of identification.
- It is hard to find a compelling instrument since they need to be relevant and exogenous.

Packages

```
library(tidyverse)
library(dslabs)
library(modelsummary)
library(broom)
library(car)
library(fixest)
```

Bad Controls:

```r
n <- 100

set.seed(1)

# Generate data in a tibble
data_sim <- tibble(
  e1 = rnorm(n, sd = 3),
  x = runif(n, min = 0, max = 10),
  y = runif(n, min = 10, max = 20),
  a = 6 + 2*x - 1.5*y + e1
)

# Example of a bad control
data_sim$c <- data_sim$x + rnorm(nrow(data_sim), sd = 0.5)
data_sim$d <- data_sim$a + rnorm(nrow(data_sim), sd = 0.5)
```

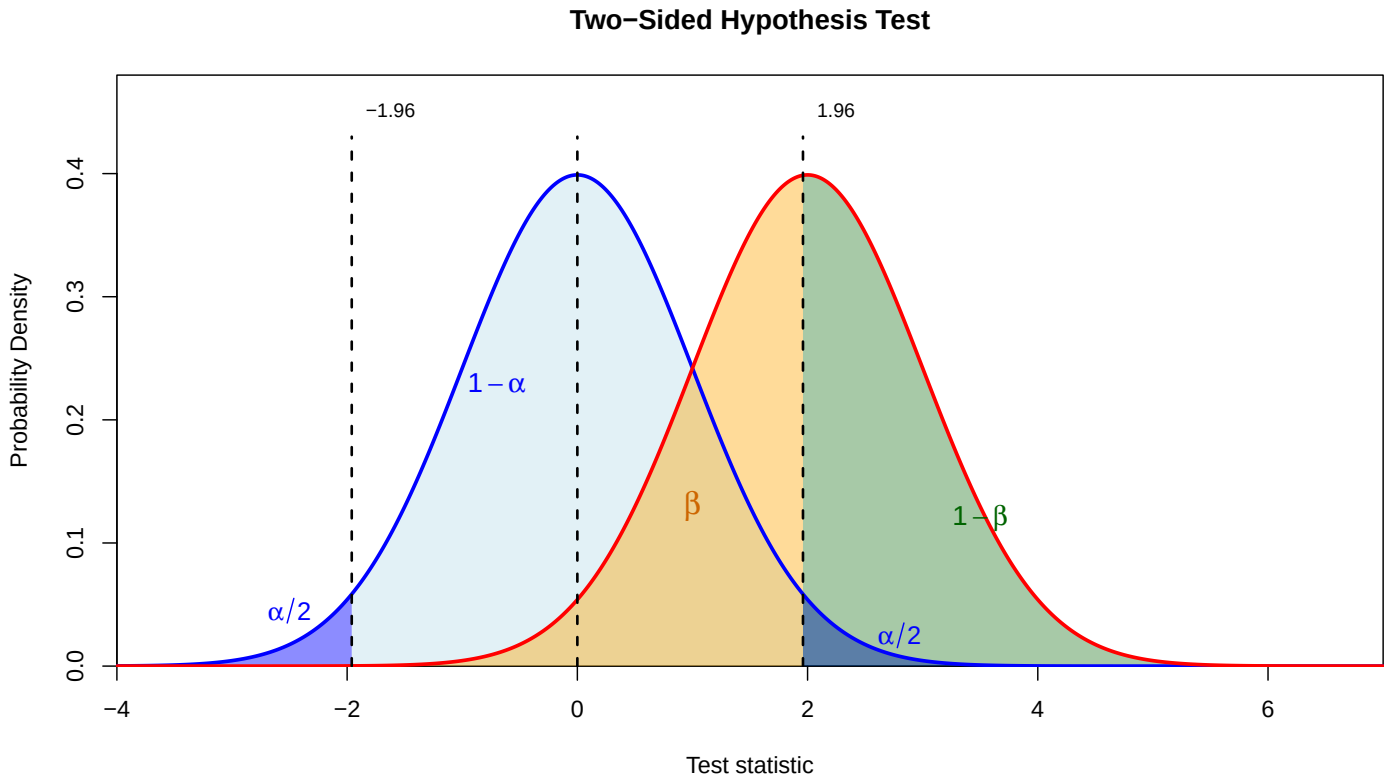
Bad Controls:

```
# Correct model
lm1 <- lm(data = data_sim, a ~ x + y)

# Bad control: creates multicollinearity
lm2 <- lm(data = data_sim, a ~ c + y + x)

# Very bad control: soaks up variation in the outcome
# and can block causal pathways
lm3 <- lm(data = data_sim, a ~ d + y + x)
```

Power, Type-1 and Type-2 Errors:



Power, Type 1 and Type 2 Errors:

Trade-off between type 1 and type 2 errors:

```
Alpha (Type I Error): 0.05
1 - Alpha (Confidence Level): 0.95
Beta (Type II Error): 0.484
1 - Beta (Statistical Power): 0.516
```

Omitted Variable Bias in OLS

Suppose the true model contains two regressors:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + u_i$$

If we exclude z_i, we estimate:

$$y_i = \alpha_0 + \alpha_1 x_i + e_i$$

Under exogeneity of the full model, the asymptotic omitted variable bias is:

$$\boxed{\operatorname{plim}(\hat{\alpha}_1) - \beta_1 = \beta_2 \frac{\operatorname{Cov}(x, z)}{\operatorname{Var}(x)}}$$

Equivalently,

$$\operatorname{plim}(\hat{\alpha}_1) = \beta_1 + \beta_2 \frac{\operatorname{Cov}(x, z)}{\operatorname{Var}(x)}$$

Addressing Omitted Variable Bias: Controls and IV

1. OLS with controls

Include the omitted variable z:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + u_i$$

Under the conditional exogeneity assumption:

$$E[u_i \mid x_i, z_i] = 0$$

2. Instrumental variables (IV)

Use an instrument w for the endogenous regressor x.

- **Relevance:** $\mathrm{Cov}(w, x) \neq 0$
- **Instrument exogeneity:** $\mathrm{Cov}(w, u) = 0$
- **Exclusion restriction:** w affects y only through x

Instrumental Variables (IV)

Consider the structural model:

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

The IV estimator with one instrument and one endogenous regressor is:

$$\boxed{\hat{\beta}_{IV} = \frac{\widehat{\mathrm{Cov}}(w, y)}{\widehat{\mathrm{Cov}}(w, x)}}$$

The estimator is consistent when the instrument satisfies the relevance, exogeneity, and exclusion restrictions.